Luleå University of Technology

[Department of Civil, Environmental and Natural Resources Engineering]

Course: D7015B – Industrial AI & eMaintenance

Assignment 5

“Point Cloud Processing Techniques”

Submitted by:
Akbar Ali Khan
Student ID: 1158006726

Supervisor: Amit Patwardhan

Date 18 March 2026

Table of Contents

Objective 3

Dataset Description 3

Work Process & Observations 3

- **1. Ground Level Estimation** 3
- **2. DBSCAN Clustering** 4
- **3. Clustering Results** 5
- **4. Catenary Identification** 6

Code 8

Conclusion 10

GitHub Repository 10

Objective

The objective of this assignment is to process LiDAR point cloud data using Python and identify structures present in the given dataset. The main focus is on estimating the ground level, applying clustering techniques and extracting the catenary structure.

Dataset Description

Following datasets were provided for this assignment:

- Dataset 1
- Dataset 2

Each dataset contains 3D point cloud with X, Y and Z coordinates. These points represent different elements of railway infrastructure.

Work Process & Observations

1. Ground Level Estimation

The ground level was estimated by plotting a histogram of the Z values. The most frequent value was selected as the ground level.

Results:

- Dataset 1: Ground ≈ 61.25
- Dataset 2: Ground ≈ 61.27

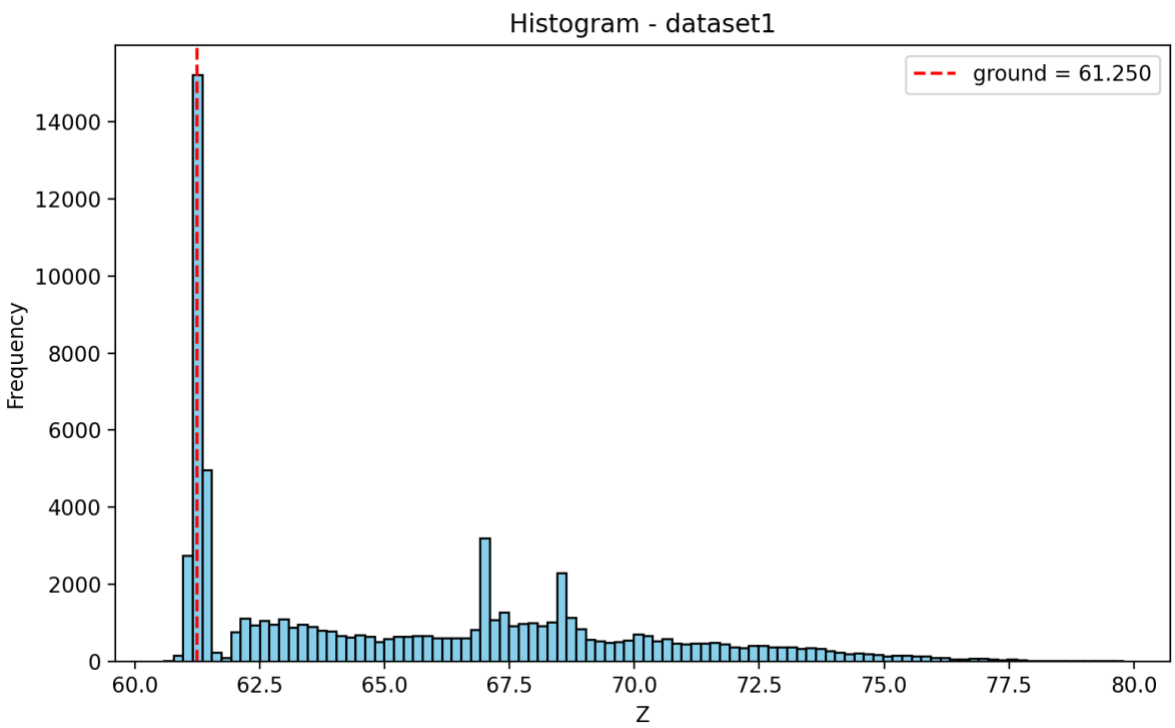

Figure 1: Histogram of Z values (Dataset 1)

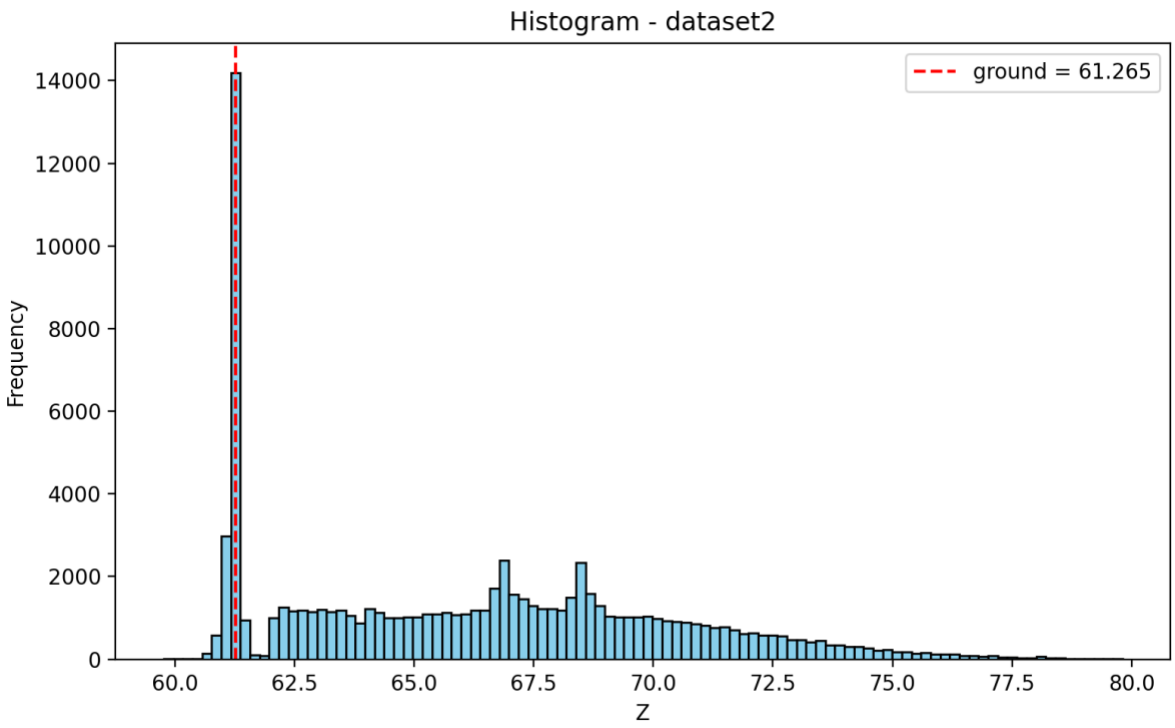

Figure 2: Histogram of Z values (Dataset 2)

Observation:

The ground level values for both datasets are very close, which shows consistency in the data.

2. DBSCAN Clustering

To cluster the data, DBSCAN was used. The value of eps was selected using the elbow method based on k-nearest neighbour distances.

From the elbow plot, the selected value was:

- eps = 0.5

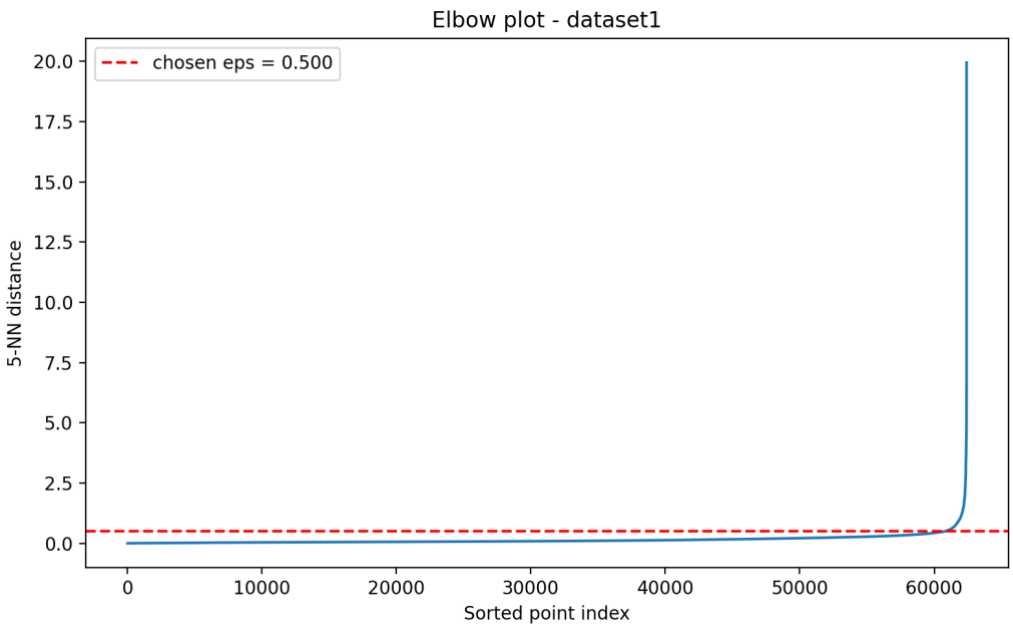

Figure 3: Elbow Plot (Dataset 1)

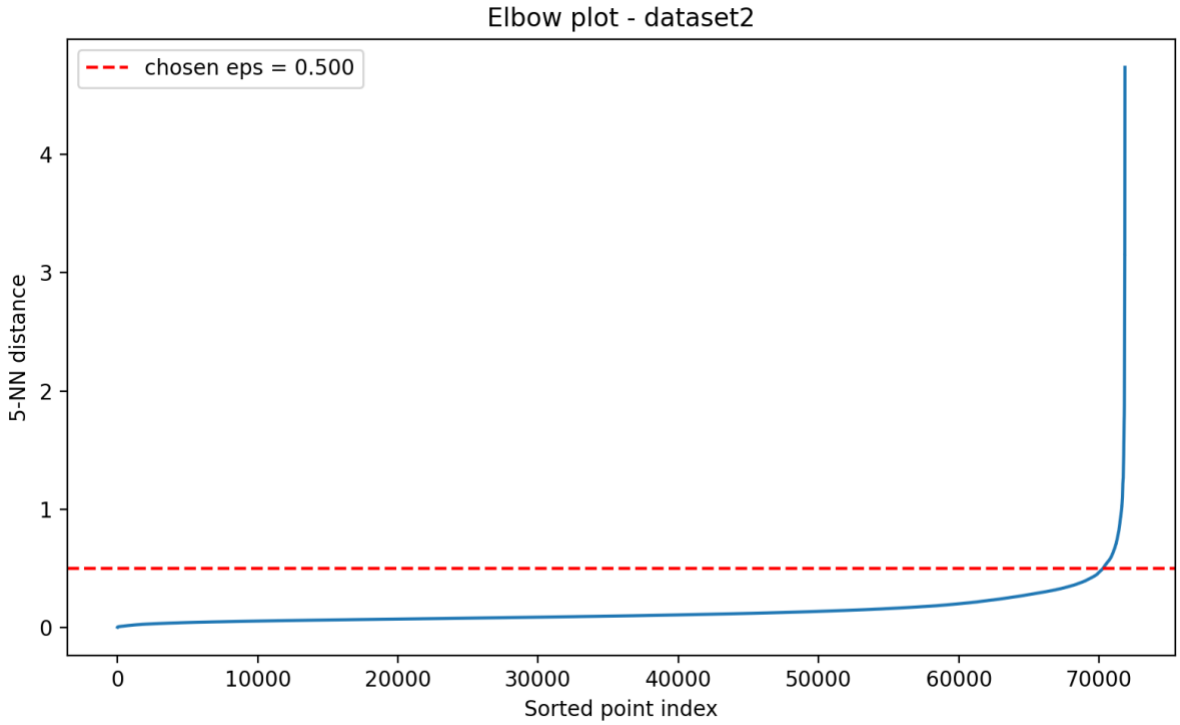

Figure 4: Elbow Plot (Dataset 2)

Observation:

This value gave a good balance between separating clusters and avoiding too much noise. Lower values resulted in fragmented clusters while higher values merged different structures.

3. Clustering Results

DBSCAN was applied using the selected eps value to group the points.

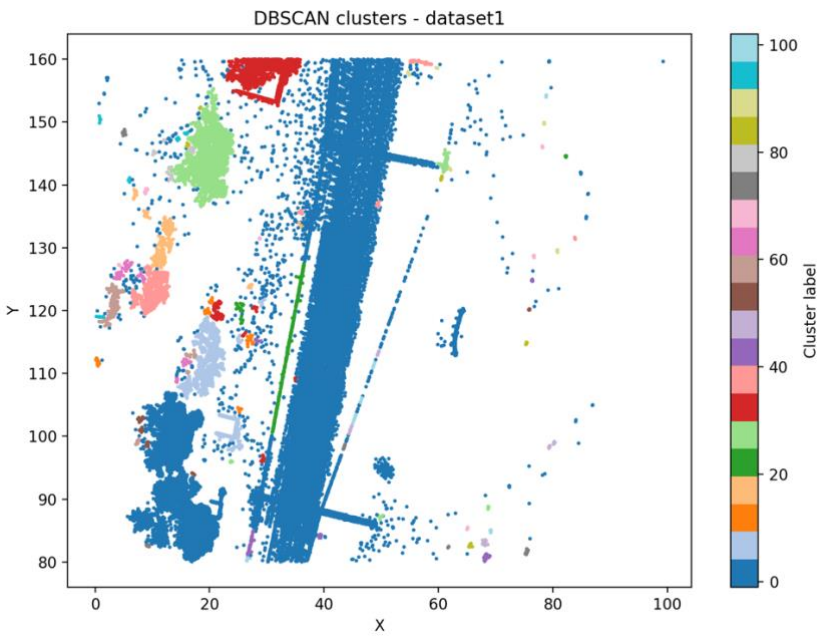

Figure 5: DBSCAN clustering (Dataset 1)

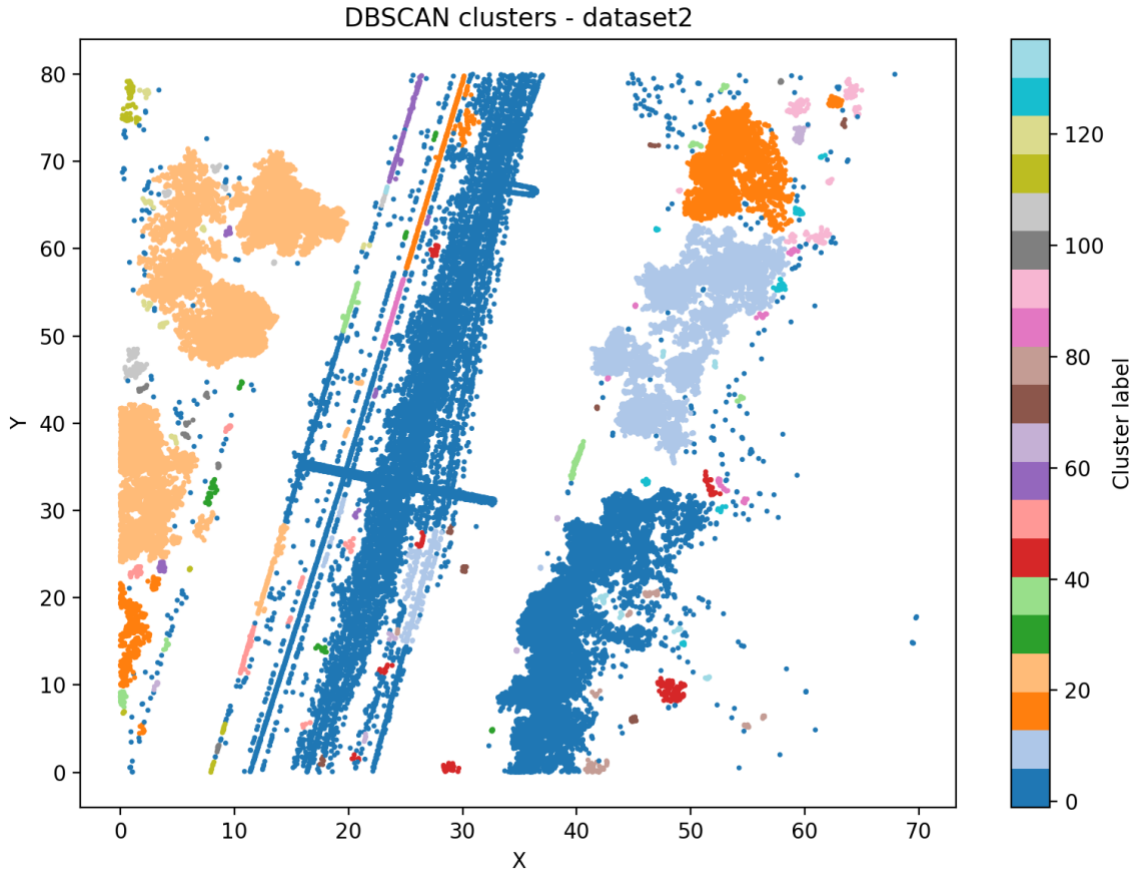

Figure 6: DBSCAN clustering (Dataset 2)

Observation:

Different clusters representing various structures can be seen clearly. Some noise points are also present in the plots.

4. Catenary Identification

The largest cluster (excluding noise) was assumed to represent the catenary.

To verify this, the spread of the cluster in X and Y directions was analysed.

Results:

Dataset 1:

- min(x): 27.08
- max(x): 59.88
- min(y): 80.01
- max(y): 160.00

Dataset 2:

- min(x): 11.39
- max(x): 37.01

- min(y): 0.04
- max(y): 79.99

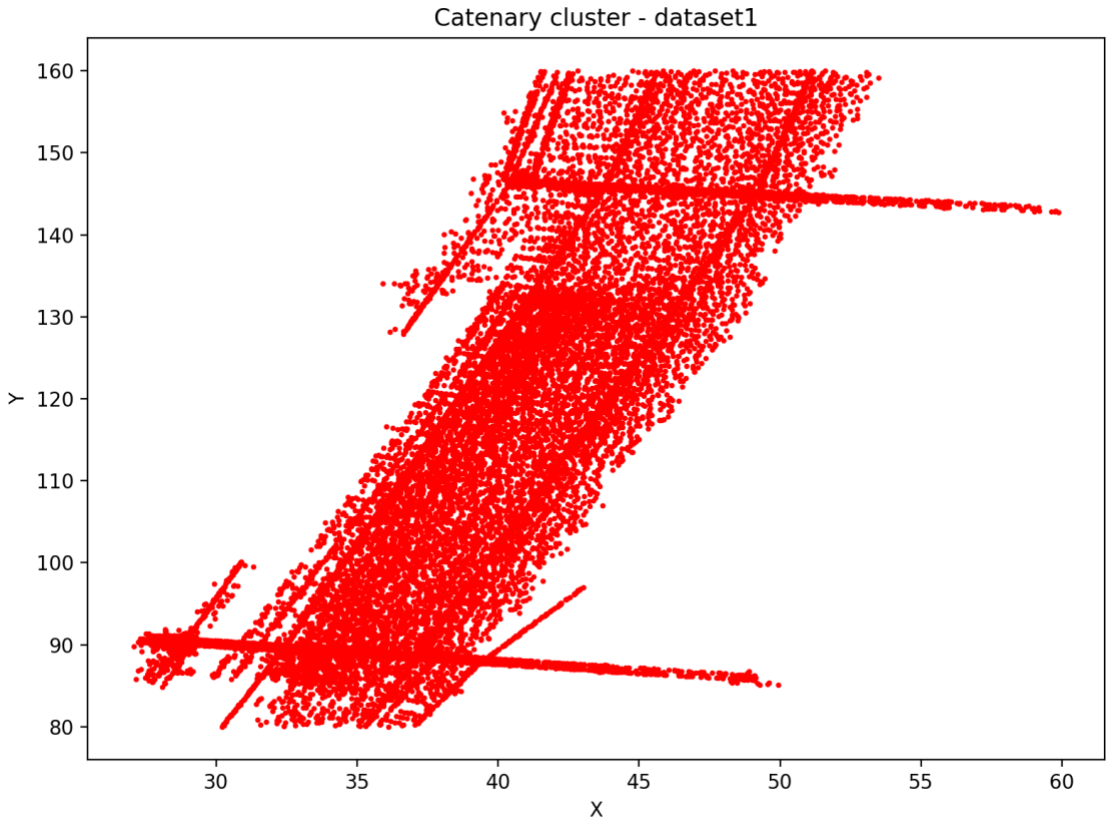

Figure 7: Catenary cluster (Dataset 1)

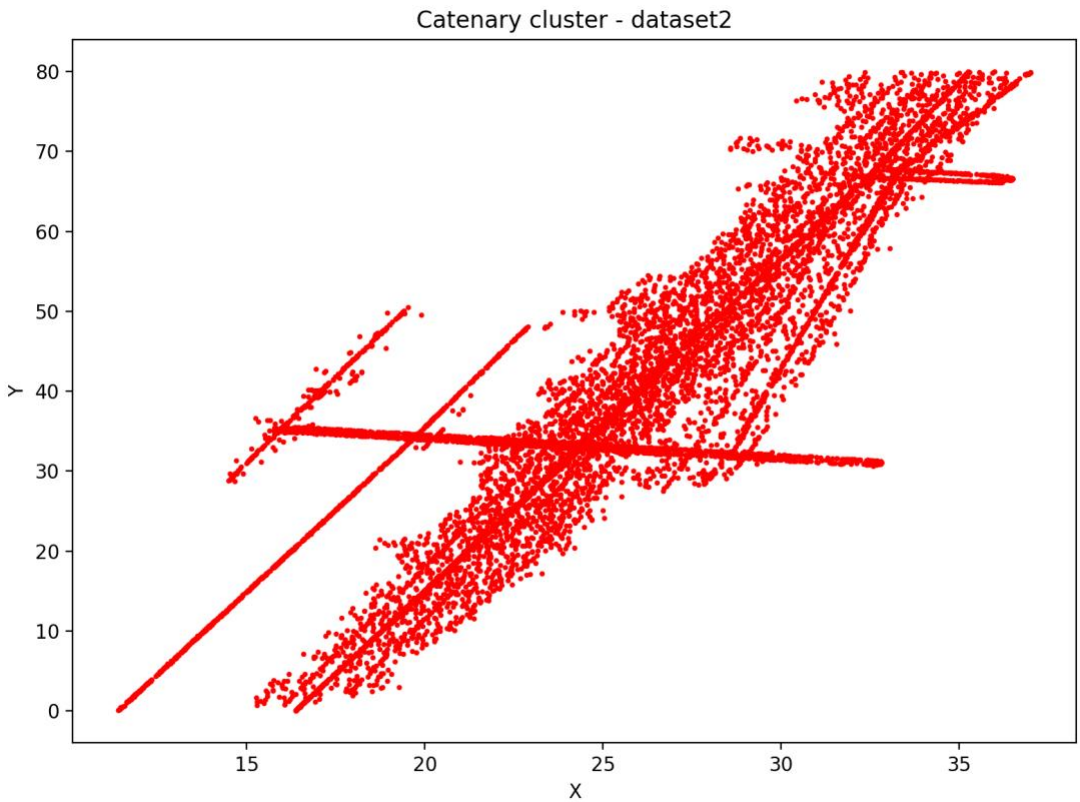

Figure 8: Catenary cluster (Dataset 2)

Observation:

The selected cluster shows a continuous and elongated shape which is consistent with the expected structure of a catenary.

Code

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import DBSCAN
from sklearn.neighbors import NearestNeighbors
import os

files = ["dataset1.npy", "dataset2.npy"]

if not os.path.exists("images"):
    os.makedirs("images")

for file in files:
    name = file.replace(".npy", "")
    data = np.load(file)

    print("\nWorking on", name)
    print("Shape:", data.shape)

    x = data[:, 0]
    y = data[:, 1]
    z = data[:, 2]

    # Task 1 - ground level
    counts, bins = np.histogram(z, bins=100)
    idx = np.argmax(counts)
    ground_level = (bins[idx] + bins[idx + 1]) / 2

    print("Ground level:", ground_level)

    plt.figure(figsize=(8, 5))
    plt.hist(z, bins=100, color="skyblue", edgecolor="black")
    plt.axvline(ground_level, color="red", linestyle="--", label=f"ground =
{ground_level:.3f}")
    plt.xlabel("Z")
    plt.ylabel("Frequency")
    plt.title(f"Histogram - {name}")
    plt.legend()
    plt.tight_layout()
    plt.savefig(f"images/{name}_histogram.png", dpi=200)
    plt.show()

    # keep points above ground
    data2 = data[data[:, 2] > ground_level]
    xy = data2[:, :2]

    print("Points above ground:", data2.shape)

    # Task 2 - elbow plot only for visual support
    nbrs = NearestNeighbors(n_neighbors=5)
    nbrs.fit(xy)
    distances, indices = nbrs.kneighbors(xy)

    k_dist = np.sort(distances[:, 4])
```

```
# chosen eps manually
if name == "dataset1":
    eps = 0.5
else:
    eps = 0.5

print("Chosen eps:", eps)

plt.figure(figsize=(8, 5))
plt.plot(k_dist)
plt.axhline(eps, color="red", linestyle="--", label=f"chosen eps = {eps:.3f}")
plt.xlabel("Sorted point index")
plt.ylabel("5-NN distance")
plt.title(f"Elbow plot - {name}")
plt.legend()
plt.tight_layout()
plt.savefig(f"images/{name}_elbow.png", dpi=200)
plt.show()

# DBSCAN
db = DBSCAN(eps=eps, min_samples=5)
labels = db.fit_predict(xy)

unique_labels = np.unique(labels)
n_clusters = len(unique_labels) - (1 if -1 in unique_labels else 0)
print("Clusters found:", n_clusters)

plt.figure(figsize=(8, 6))
plt.scatter(data2[:, 0], data2[:, 1], c=labels, cmap="tab20", s=2)
plt.colorbar(label="Cluster label")
plt.xlabel("X")
plt.ylabel("Y")
plt.title(f"DBSCAN clusters - {name}")
plt.tight_layout()
plt.savefig(f"images/{name}_clusters.png", dpi=200)
plt.show()

# Task 3 - find largest cluster using x,y span
best_label = None
best_points = None
best_span = -1

for lab in unique_labels:
    if lab == -1:
        continue

    pts = data2[labels == lab]

    if len(pts) == 0:
        continue

    span_x = pts[:, 0].max() - pts[:, 0].min()
    span_y = pts[:, 1].max() - pts[:, 1].min()
    total_span = span_x + span_y

    if total_span > best_span:
        best_span = total_span
        best_label = lab
        best_points = pts

if best_points is not None:
```

```
        min_x = best_points[:, 0].min()
        min_y = best_points[:, 1].min()
        max_x = best_points[:, 0].max()
        max_y = best_points[:, 1].max()

        print("Catenary cluster label:", best_label)
        print("min(x):", min_x)
        print("min(y):", min_y)
        print("max(x):", max_x)
        print("max(y):", max_y)

        plt.figure(figsize=(8, 6))
        plt.scatter(best_points[:, 0], best_points[:, 1], color="red", s=3)
        plt.xlabel("X")
        plt.ylabel("Y")
        plt.title(f"Catenary cluster - {name}")
        plt.tight_layout()
        plt.savefig(f"images/{name}_catenary.png", dpi=200)
        plt.show()
    else:
        print("No valid cluster found")
```

Conclusion

In this assignment, LiDAR data was processed to identify structures using clustering techniques. The ground level was estimated using histogram analysis and DBSCAN was used to group the data points.

The method worked well in separating different structures and the catenary was successfully identified as the largest cluster in both given datasets. The results were consistent across both datasets.

GitHub Repository

https://github.com/uckbarr-cloud/assignment5-lidar-processing